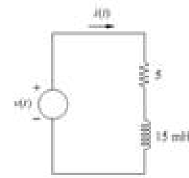
Exercise 12

- A non-sinusoidal voltage source has a fourier series of

$$v(t) = 10 + 20\cos(2\pi 60t - 25^\circ) + 30\cos(4\pi 60t + 20^\circ)\ \text{V}$$

- This voltage is connected to a load that is 5 ohm resistor and 15 mH inductor in series.
- Determine the power absorbed by the load.



Solution

- The dc term is

$$I_0 = \frac{V_0}{R} = \frac{10}{5} = 2\ \text{A}$$

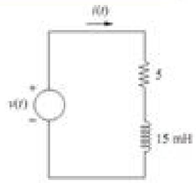
- AC current terms are computed from phasor analysis:

$$I_1 = \frac{V_1}{R + j\omega_1 L} = \frac{20\angle(-25^\circ)}{5 + j(2\pi 60)(0.015)} = 2.65\angle(-73.5^\circ)\ \text{A}$$

$$I_2 = \frac{V_2}{R + j\omega_2 L} = \frac{30\angle 20^\circ}{5 + j(4\pi 60)(0.015)} = 2.43\angle(-46.2^\circ)\ \text{A}$$

- Load current then can be calculated as

$$i(t) = 2 + 2.65\cos(2\pi 60t - 73.5^\circ) + 2.43\cos(4\pi 60t - 46.2^\circ)\ \text{A}$$



Solution (cont.)

- The power at each frequency in the Fourier series can be determined as follows,

dc term: $P_0 = (10\ \text{V})(2\ \text{A}) = 20\ \text{W}$

$\omega = 2\pi 60$: $P_1 = \frac{(20)(2.65)}{2}\cos(-25^\circ + 73.5^\circ) = 17.4\ \text{W}$

$\omega = 4\pi 60$: $P_2 = \frac{(30)(2.43)}{2}\cos(20^\circ + 46^\circ) = 14.8\ \text{W}$

- Total power is then

$$P = 20 + 17.4 + 14.8 = 52.2\ \text{W}$$

- **Alternative Method:** Since the average power of inductor is zero, the power absorbed by the load can be calculated using rms current as follows

$$P = I_{rms}^2 R = \left[2^2 + \left(\frac{2.65}{\sqrt{2}}\right)^2 + \left(\frac{2.43}{\sqrt{2}}\right)^2\right] 5 = 52.2\ \text{W}$$